

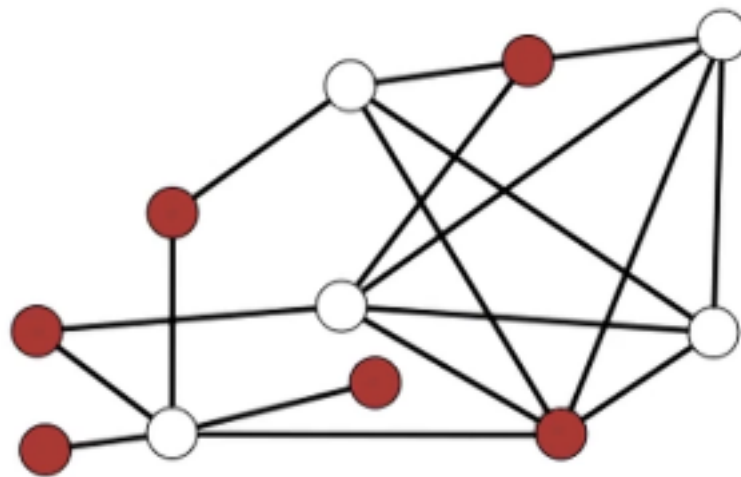
NP3: Graph Problems

Graph Problems Lecture Outline I

1. 3SAT is NP-complete
2. Upcoming:
 - Independent Sets
 - Clique
 - Vertex Cover

Independent Set

1. For undirected $G=(V,E)$, a subset S in V is an independent set if no edges are contained in S
 - For all x,y in S , (x,y) is not in E



Independent Set

Quiz: Max Independent Set

1. The max independent set problem is known to be in NP
 - False; for a given problem to be in NP, a given solution must be verifiable in polynomial time. Max IS is not verifiable in polynomial-time.

Search Version

1. Input: Undirected $G=(V,E)$ and goal g
2. Output: Independent set S with size $|S| \geq g$ if one exists and NO otherwise
3. Theorem: The independent set problem is NP-complete

Proof Outline

1. The independent set problem is NP-complete
 - Need to show independent set is in NP
 - Given input G and goal g and solution S
 - In $O(n^2)$ time can check all pairs x,y in S : verify (x,y) not in E
 - In $O(n)$ time check $|S| \geq g$

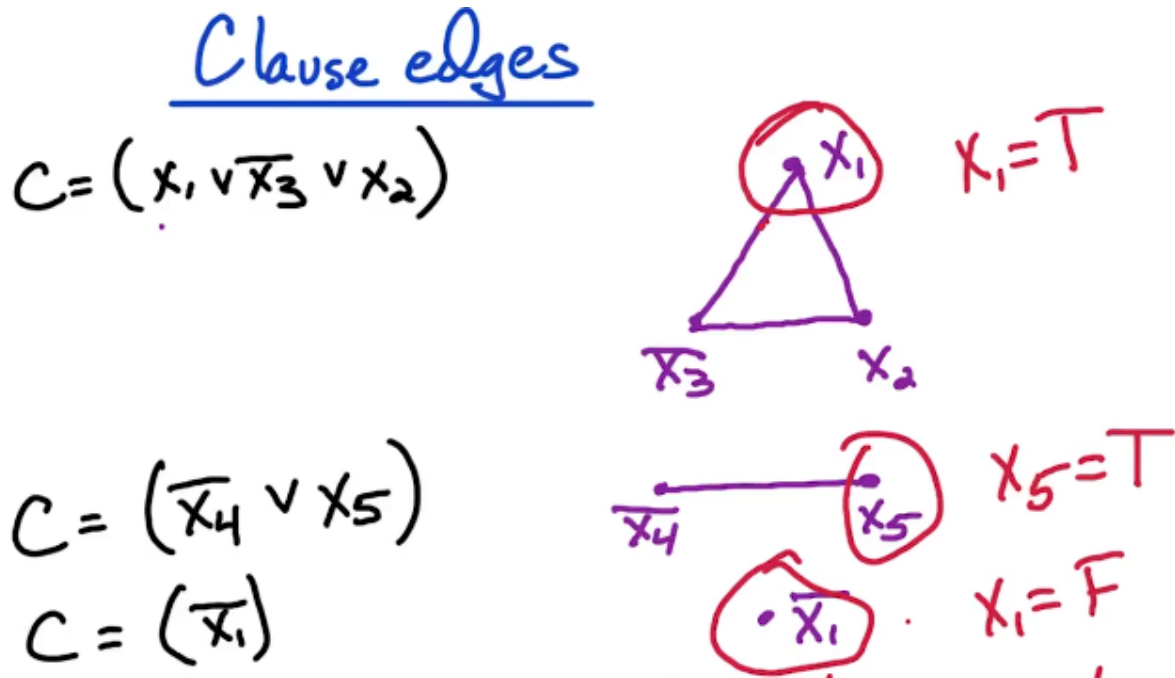
- Show 3SAT can be reduced to independent set
 - 3SAT is simpler than SAT

3SAT IS

1. Consider 3SAT input f with variables $x_1, \dots, x_n$ and clauses $C_1, \dots, C_m$
 - Each clause has size $|C_i| \leq 3$
2. We'll define a graph G and set $g = m$ (number of clauses)
 - Idea: For each clause C_i , create $|C_i|$ vertices

Clause Edges

1. Clause $C = (x_1 \vee \neg x_3 \vee x_2)$
 - Add edges between all variables in the clause
2. Independent set S has ≤ 1 vertex per clause
3. Since $g = m$, solution has $= 1$ vertex per clause



Clause Edges

Variable Edges

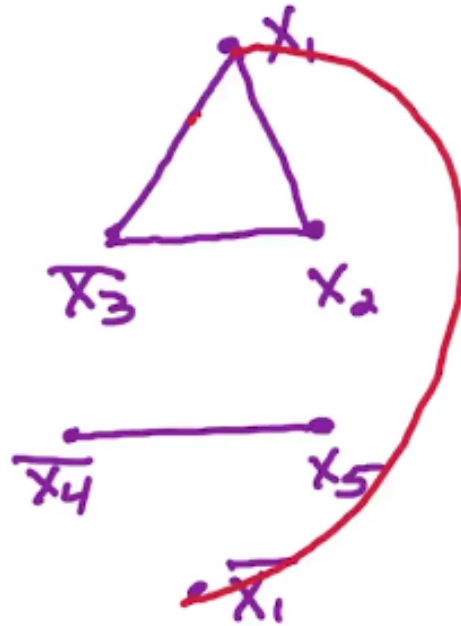
1. For each x_i : Add edges between all x_i and all $\neg x_i$

Variable edges

$$C = (x_1 \vee \overline{x}_3 \vee x_2)$$

$$C = (\overline{x}_4 \vee x_5)$$

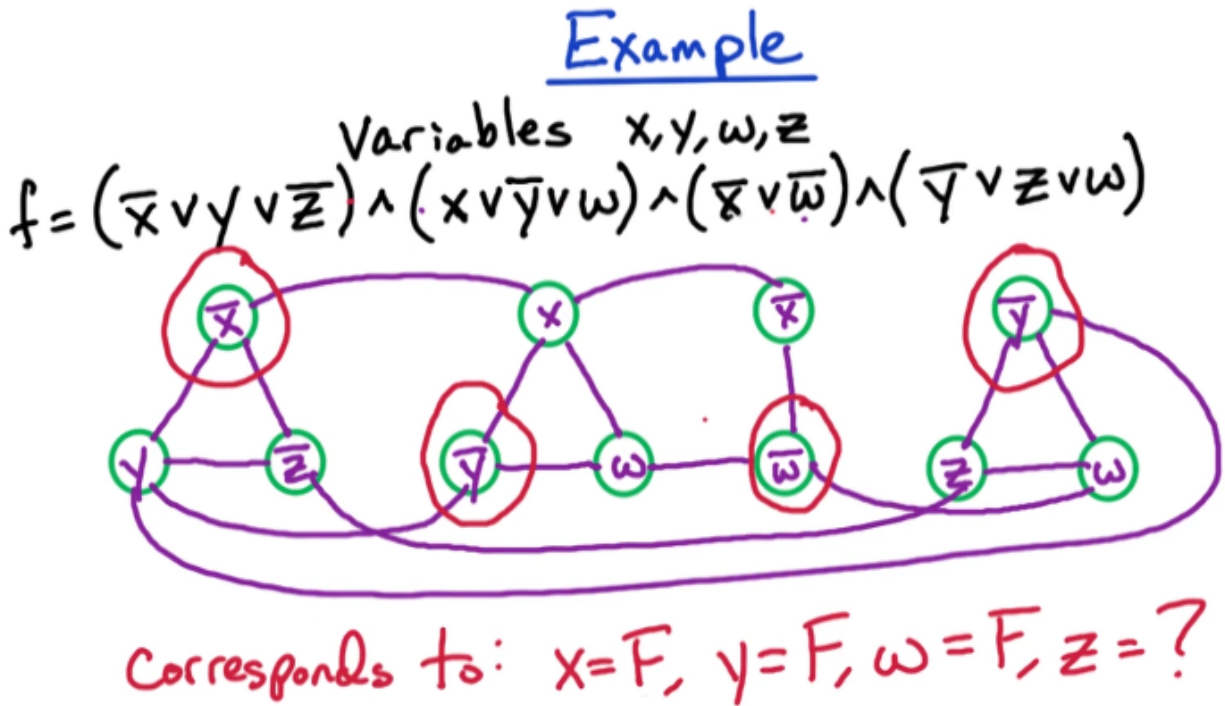
$$C = (\overline{x}_1)$$



Variable Edges

Example

- Variables x, y, w, z
 - $f = (!x \vee y \vee !z) \wedge (x \vee !y \vee w) \wedge (!x \vee !w) \wedge (!y \vee z \vee w)$



Example

Graph Problems Correctness

1. f has a satisfying assignment if and only if G has an independent set of size $\geq g$
 - Consider a satisfying assignment for f
 - For each clause C , take 1 of the satisfied literals
 - * Add corresponding vertex to S
 - * $|S| = m = g$
 - S has =1 vertex per clause and not both x_i and $\neg x_i$ because it corresponds to a satisfying assignment (can't have both)
 - * No clause edges and no variable edges

Reverse Implication

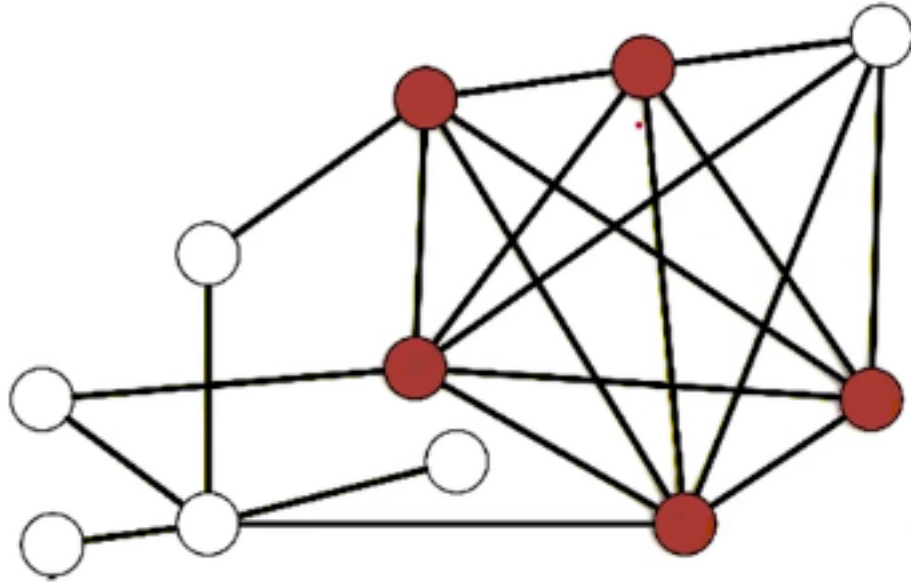
1. f has a satisfying assignment if and only if G has an independent set of size $\geq g$
 - Take independent set S of size $\geq g$ has =1 vertex per clause
 - Set corresponding literal to True $\Rightarrow$ every clause is satisfied
 - * No contradictory literals since edges $x_i \leftrightarrow \neg x_i$, so valid assignment

NP-hard

1. Max Independent Set Problem
 - IS (search) is in NP-complete
 - Max-IS
 - IS (size at least g) $\rightarrow$ Max-IS (max)
 - * Straightforward to reduce the search version to the optimization version
 - Max-IS is at least as hard as every problem in NP
 - Theorem: Max-Independent Set problem is NP-hard

Clique

1. Clique = Fully connected subgraph
 - For $G=(V,E)$, subset S is a clique if for all x,y in S , (x,y) is in E
2. Want to find large cliques



Clique

Clique: Search Version

1. Input: $G=(V,E)$ and goal g
2. Output: S in V where S is a clique of size $|S| \geq g$ if one exists, NO otherwise
3. Theorem: Clique problem is NP-complete

Clique: Proof Outline

1. Clique is in NP
 - Given input (G,g) and S
 - For all x,y in S , check that (x,y) in E in $O(n^2)$
 - Check that $|S| \geq g$ in $O(n)$
 - Show that clique is at least as hard as every problem in NP-complete
 - Reduce IS to clique

IS Clique Idea

1. Key idea: Clique is opposite of Independent Set
 - Clique is fully connected (all edges within S)
 - Independent set has no edges within S
2. For $G=(V,E)$, let $!G = (V,!E)$ where:
 - $!E = \{(x,y): (x,y) \text{ not in } E\}$
 - (x,y) in $!E \iff (x,y)$ not in E
3. Observation:
 - S is a clique in $!G \iff S$ is an independent set in G

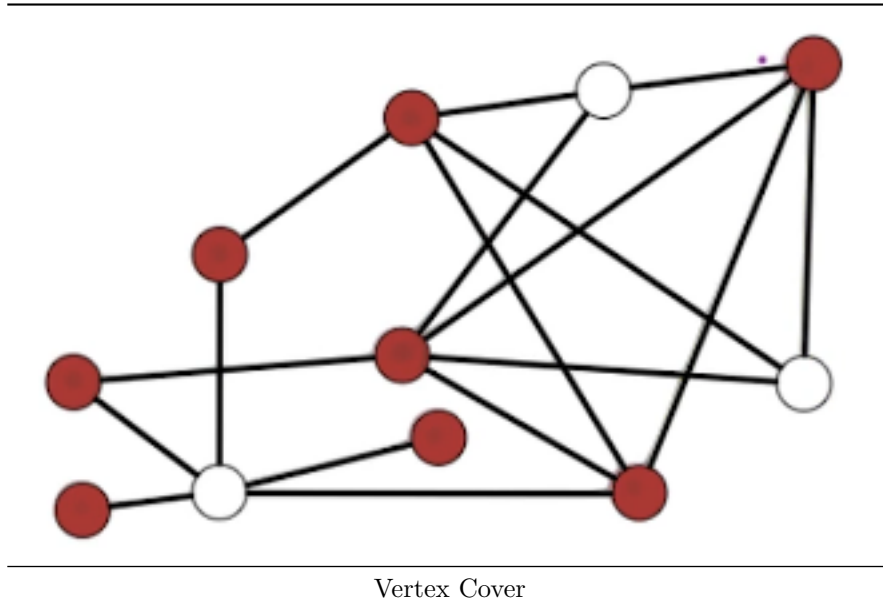
IS Clique

1. Independent Set $\rightarrow$ Clique:

- Given input $G=(V,E)$ and goal g for IS problem, let $\neg G$ and g be input to the clique problem
- If we get solution S for clique, then return S for IS problem
- If we get NO, then return NO

Vertex Cover

1. For $G=(V,E)$, a subset of vertices S is a vertex cover if S “covers every edge”
 - For every (x,y) in E , either x is in S and/or y is in S



VC: Search Version

1. Input: $G=(V,E)$ and budget b (trying to find minimum vertex cover)
2. Output: Vertex cover S of size $|S| \leq b$ if one exists and NO otherwise
3. Theorem: Vertex cover problem is NP-complete

VC: Proof Outline

1. VC is in NP
 - Given input (G,b) and proposed solution S
 - For every (x,y) in E , ≥ 1 of x or y are in S in $O(n+m)$ time
 - Check that $|S| \leq b$ in $O(n)$ time
 - Reduce IS to VC

VC: Reduction Idea

1. Claim: S is a vertex cover $\iff \neg S$ is an independent set

Forward Implication

1. Take vertex cover S
 - For edge (x,y) in E : ≥ 1 of x or y are in S
 - Therefore, ≤ 1 of x or y in $\neg S$

- Therefore, no edge contained in I , so I is an independent set

Reverse Implication

1. Take independent set I
 - For every (x,y) in E , ≤ 1 of x or y in I
 - Therefore, ≥ 1 of x or y in S
 - Therefore, S covers every edge

IS VC

1. For input $G=(V,E)$ and g for independent set
 - Let $b = n - g$
 - Run vertex cover on G, b
2. G has a vertex cover of size $\leq n - g \iff G$ has an independent set of size $\geq g$

IS VC Correctness

1. (G,g) for IS $\rightarrow (G,b)$ for VC
2. Given solution S for VC return I as solution to IS problem
3. If NO solution for VC, return NO for IS problem

Practice Problems

1. DPV 8.4: NP-completeness error
2. DPV 8.10: Proof by generalization
3. DPV 8.14: Clique + IS
4. 8.19: Kite